

OnlyFunds App

A Report submitted under Project-Based
Learning In Partial Fulfillment of the Course Requirement
for
“Web Technologies(22IT104001)”

Submitted By

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DREAM. BELIEVE. ACHIEVE

Department of Computer Science and Engineering
School of Computing

MOHAN BABU UNIVERSITY

Sree Sainath Nagar, Tirupati -517102

2025-2026



MOHANBABUUNIVERSITY

Vision

To be a globally respected institution with an innovative and entrepreneurial culture that offers transformative education to advance sustainability and societal good.

Mission

- ❖ Develop industry-focused professionals with a global perspective.
- ❖ Offer academic programs that provide transformative learning experience founded on the spirit of curiosity, innovation, and integrity.
- ❖ Create confluence of research, innovation, and idea to bring about sustainable and socially relevant enterprises.
- ❖ Uphold high standards of professional ethics leading to harmonious relationship with environment and society.

SCHOOL OF COMPUTING

Vision

To lead the advancement of computer science research and education that has real-world impact and to push the frontiers of innovation in the field.

Mission

- ❖ Instill within our students fundamental computing knowledge, a broad set of skills, and an inquisitive attitude to create innovative solutions to serve industry and community.
- ❖ Provide an experience par excellence with our state-of-the-art research, innovation, and incubation ecosystem to realise our learners' fullest potential.
- ❖ Impart continued education and research support to working professionals in the computing domain to enhance their expertise in the cutting-edge technologies.

- ❖ Inculcate among the computing engineers of tomorrow with a spirit to solve societal challenges.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Vision

To become a Centre of Excellence in Computer Science and its emerging areas by imparting high quality education through teaching, training and research.

Mission

- Imparting quality education in Computer Science and Engineering and emerging areas of IT industry by disseminating knowledge through contemporary curriculum, competent faculty and effective teaching-learning methodologies.
- Nurture research, innovation and entrepreneurial skills among faculty and students to contribute to the needs of industry and society.
- Inculcate professional attitude, ethical and social responsibilities for prospective and promising engineering profession.
- Encourage students to engage in life-long learning by creating awareness of the contemporary developments in Computer Science and Engineering and its emerging areas.

B.Tech.Computer Science and Engineering

PROGRAM EDUCATIONAL OBJECTIVES

After few years of graduation, the graduates of B.Tech.CSE will be:

- PEO1.**Pursuing higher studies in core, specialized or allied areas of Computer Science, or Management.
- PEO2.**Employed in reputed Computer and I.T organizations or Government to have a globally competent professional career in Computer Science and Engineering domain or be successful Entrepreneurs.
- PEO3.** Able to demonstrate effective communication, engage in teamwork, exhibit leadership skills and ethical attitude, and achieve professional advancement through continuing education.

PROGRAM OUTCOMES

On successful completion of the Program,the graduates of B.Tech.CSE Program will be able to:

- PO1.Engineering Knowledge** :Apply the knowledge of mathematics ,science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2.Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4.Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments ,analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5.Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

- PO6.The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7. Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8.Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9.Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10.Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11.Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12.Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

On successful completion of the Program ,the graduates of B.Tech.(CSE)program will be able to:

- PSO1.** Apply knowledge of computer science engineering, Use modern tools, techniques and technologies for efficient design and development of computer-based systems for complex engineering problems.
- PSO2.**Design and deploy networked systems using standards and principles, evaluate security measures for complex networks, apply procedures and tools to solve networking issues.
- PSO3.** Develop intelligent systems by applying adaptive algorithms and methodologies for solving problems from inter-disciplinary domains.
- PSO4.**Apply suitable models ,tools and techniques to perform data analytics for effective decision making.

Course Code
22IT104001

Course Title
WEB TECHNOLOGIES

L T P S C
3 - 2 4 5

COURSE OUTCOMES: After successful completion of this course, the students will be able to:

CO1. Demonstrate knowledge on webpage design elements, dynamic content and database connection.

CO2. Analyze user requirements to develop web applications.

CO3. Design client-server applications using web technologies.

CO4. Demonstrate problem-solving skills to develop enterprise web applications.

CO5. Apply HTML, CSS, JavaScript, JQuery, Bootstrap and PHP technologies for device independent web application development.

CO6. Apply web technologies to develop interactive, dynamic and scalable web applications for societal needs.

CO-PO-PSO Mapping Table:

Course Outcomes	Program Outcomes												Program Specific Outcomes			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	-	-	-	-	-	-	-	-	-	3	2	3	-
CO2	3	3	3	2	2	-	-	-	-	-	-	-	3	2	3	-
CO3	3	3	3	2	2	-	-	-	-	-	-	-	3	2	3	-
CO4	3	3	3	2	2	-	-	-	-	-	-	-	2	2	3	-
CO5	3	2	2	2	2	3	-	-	-	-	-	-	2	2	3	-
CO6				2					3	3						
Course Correlation Mapping	3	3	3	2	2				3	3			3	2	3	

Correlation Levels: 3: High; 2: Medium; 1: Low



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Sree Sainath Nagar, Tirupati 517 102

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Project Entitled

“OnlyFunds App”

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Is the work submitted under Project-Based Learning in Partial Fulfillment of the Course Requirements for “Web Technologies(22IT104001)” during 2025-2026.

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OnlyFunds Application - Complete Technical Documentation

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1. Introduction

1.1 Project Overview

OnlyFunds is a modern web-based crowdfunding platform designed to connect individuals seeking financial support with potential donors. The application facilitates transparent fundraising through user-created fund request posts, secure donation processing, and social interaction features including comments and direct messaging.

1.2 Purpose and Objectives

The primary purpose of OnlyFunds is to provide an accessible platform where users can create fund requests for various needs—whether for medical expenses, education, creative projects, or community initiatives. The platform aims to foster trust and community engagement through transparent communication and user verification.

Key Objectives:

- Enable users to create and manage fundraising campaigns
- Facilitate secure and trackable donations
- Build community through social features like comments and messaging
- Ensure platform security and user privacy
- Provide an intuitive and accessible user experience

1.3 Target Audience

This documentation is intended for:

- **Developers** working on maintaining or extending the application
- **Project Managers** overseeing application development and deployment
- **System Administrators** managing deployment and infrastructure
- **End Users** seeking to understand platform capabilities
- **Stakeholders** evaluating the application architecture and design

1.4 Document Structure

This documentation follows a structured approach, beginning with high-level overviews and progressing to detailed technical specifications. Each section is designed to be self-contained while building upon previous content for comprehensive understanding.

2. Project Overview

2.1 Background

OnlyFunds was developed to address the growing need for accessible crowdfunding platforms that prioritize transparency and community engagement. Traditional fundraising methods often lack the reach and efficiency provided by digital platforms, while existing solutions may be complex or inaccessible to non-technical users.

2.2 Problem Statement

Many individuals and organizations struggle to raise funds due to:

- Limited access to traditional fundraising platforms
- High platform fees reducing actual funds received
- Lack of community engagement and trust-building features
- Complex user interfaces deterring potential donors and recipients
- Insufficient transparency in fund tracking and usage

2.3 Solution Overview

OnlyFunds addresses these challenges through:

- **Zero-cost platform access** with optional premium features
- **Social networking elements** that build trust and community
- **Transparent tracking** of fund requests and donations

- **Simple, intuitive interface** accessible to all user demographics
- **Real-time communication** through integrated chat functionality
- **Robust security** protecting user data and transactions

2.4 Key Features Summary

- User authentication and profile management
 - Fund request creation with rich media support
 - Donation tracking and management
 - Social interaction through comments and likes
 - Direct messaging between users
 - User search and discovery
 - Admin moderation and controls
 - Real-time updates and notifications
-

3. System Architecture

3.1 High-Level Architecture

OnlyFunds follows a modern serverless architecture leveraging Firebase services. The application consists of:

- **Frontend:** Next.js-based React application
- **Backend:** Firebase services (Firestore, Authentication, Hosting)
- **Database:** Cloud Firestore NoSQL database
- **Authentication:** Firebase Authentication
- **Hosting:** Firebase Hosting with CDN

3.2 Technology Stack

Frontend Technologies:

- Next.js 14+ (React framework)
- TypeScript for type safety

- Tailwind CSS for styling
- React hooks for state management

Backend Services:

- Firebase Firestore (database)
- Firebase Authentication (user management)
- Firebase Hosting (static hosting and CDN)
- Firebase Security Rules (access control)

Development Tools:

- Node.js and npm
- Git for version control
- Firebase CLI for deployment
- ESLint for code quality

3.3 Frontend Architecture

The frontend follows a component-based architecture with clear separation of concerns:

- **Pages:** Route-level components
- **Components:** Reusable UI elements
- **Hooks:** Custom logic and state management
- **Services:** Firebase SDK integration
- **Utils:** Helper functions and utilities

3.4 Backend Services

Firebase provides a fully managed backend infrastructure:

- **Firestore:** Real-time NoSQL database with offline support
- **Authentication:** Secure user authentication with multiple providers
- **Security Rules:** Declarative access control at the database level

- **Cloud Functions:** (Optional) Serverless functions for backend logic

3.5 Third-Party Integrations

Current integrations include:

- Firebase ecosystem services
 - Google Cloud Platform (billing and infrastructure)
 - Future integrations may include payment processors and analytics services
-

4. Features and Functionality

4.1 User Authentication

Sign Up:

Users can create accounts using email and password. The registration process collects basic information and creates a user profile in Firestore.

Sign In:

Authenticated users access the platform through secure login. Firebase Authentication manages session tokens and security.

Password Reset:

Users can reset forgotten passwords through email verification links.

4.2 User Profile Management

Each user has a profile containing:

- Display name and bio
- Profile picture
- Contact information
- Account creation date

- Activity statistics

Users can edit their own profiles while other users can view public profile information.

4.3 Fund Request Posts

Creating Posts:

Users create fund request posts with:

- Title and description
- Target amount
- Category tags
- Images or media
- Timeline/deadline

Post Visibility:

All posts are publicly visible to encourage discovery and engagement.

Post Management:

Post authors can edit or delete their posts. Admins have oversight capabilities.

4.4 Donation System

Making Donations:

Users can contribute to fund requests through the integrated donation system. Each donation is tracked with:

- Donor information
- Amount contributed
- Timestamp
- Associated post ID

Donation Tracking:

Both donors and recipients can track donation history and progress toward funding goals.

4.5 Comments and Interactions**Commenting:**

Users can comment on fund request posts to ask questions, offer support, or provide updates.

Likes/Reactions:

Users can like posts and comments to show support and engagement.

Comment Management:

Comment authors can edit or delete their comments. Post authors and admins have moderation capabilities.

4.6 Chat Messaging**Direct Messaging:**

Users can send direct messages to each other for private communication.

Chat Features:

- Real-time message delivery
- Message history
- Read receipts
- User online status

Privacy:

Chat messages are only accessible to the sender and receiver, enforced by Firestore security rules.

4.7 Search Functionality

User Search:

Authenticated users can search for other users by name or username to connect and communicate.

Post Search:

Users can search and filter fund request posts by category, keywords, or funding status.

4.8 Admin Controls

Admin Privileges:

Designated admin users have enhanced permissions for:

- User management
- Content moderation
- Reviewing reported content
- Accessing admin action logs

Admin Assignment:

Admin status is managed through the `/roles_admin` collection in Firestore.

5. Database Design

5.1 Database Overview

OnlyFunds uses Cloud Firestore, a NoSQL document database that provides:

- Real-time synchronization
- Offline support
- Scalability
- Flexible data modeling

5.2 Collections Structure

Primary Collections:

- `/users/{userId}` - User profiles
- `/posts/{postId}` - Fund request posts
- `/donations/{donationId}` - Donation records
- `/comments/{commentId}` - Comments on posts
- `/chat_messages/{messageId}` - Direct messages
- `/roles_admin/{userId}` - Admin role assignments
- `/admin_actions/{actionId}` - Admin activity logs
- `/content_reports/{reportId}` - Content moderation reports

5.3 Data Models

User Document:

```
text
{
  id: string,
  email: string,
  displayName: string,
  bio: string,
  profileImage: string,
  createdAt: timestamp,
  updatedAt: timestamp
}
```

Post Document:

```
text
{
  id: string,
  authorId: string,
  title: string,
  description: string,
```

```
targetAmount: number,  
currentAmount: number,  
category: string,  
images: array,  
createdAt: timestamp,  
updatedAt: timestamp,  
status: string  
}
```

Donation Document:

```
text  
{  
  id: string,  
  donorId: string,  
  postId: string,  
  amount: number,  
  message: string,  
  createdAt: timestamp  
}
```

Comment Document:

```
text  
{  
  id: string,  
  authorId: string,  
  postId: string,  
  content: string,  
  createdAt: timestamp,  
  updatedAt: timestamp  
}
```

Chat Message Document:

text

```
{  
  id: string,  
  senderId: string,  
  receiverId: string,  
  content: string,  
  createdAt: timestamp,  
  read: boolean  
}
```

5.4 Relationships and Indexes

Relationships:

- Posts reference users via **authorId**
- Donations reference posts via **postId** and users via **donorId**
- Comments reference posts via **postId** and users via **authorId**
- Chat messages reference users via **senderId** and **receiverId**

Indexes:

Composite indexes are created for common queries such as:

- Posts by author and creation date
- Comments by post and creation date
- Donations by post and amount

5.5 Security Rules

Firestore security rules enforce:

- User ownership for profile modifications
- Public read access for posts, donations, and comments
- Author-only write access for content
- Sender/receiver-only access for chat messages
- Admin-only access for moderation features

6. API and Data Flow

6.1 Firebase SDK Integration

The application uses Firebase JavaScript SDK for client-side operations. All database interactions occur through the Firestore client SDK with security enforced at the database level.

6.2 Authentication Flow

1. User submits credentials through the login form
2. Firebase Authentication validates credentials
3. Auth token is issued and stored in local storage
4. Token is attached to all Firestore requests
5. Security rules validate token and enforce permissions

6.3 Data Read/Write Operations

Reading Data:

- Client queries Firestore collections
- Security rules verify read permissions
- Data is returned if authorized

Writing Data:

- Client submits data with auth token
- Security rules verify write permissions
- Data is written if authorized and valid

6.4 Real-time Updates

Firestore provides real-time listeners that automatically update the UI when data changes. This enables features like:

- Live comment updates on posts
 - Real-time chat messaging
 - Instant notification of new donations
-

7. User Interface Design

7.1 Design Principles

OnlyFunds follows modern web design principles:

- **Simplicity:** Clean, uncluttered interface
- **Consistency:** Uniform design language throughout
- **Responsiveness:** Adapts to all screen sizes
- **Accessibility:** Meets WCAG 2.1 guidelines

7.2 Color Scheme and Typography

Primary Colors:

- Brand color: Custom gradient (orange to pink)
- Background: Dark theme with light text
- Accents: Vibrant colors for CTAs

Typography:

- Sans-serif fonts for readability
- Clear hierarchy with varied font sizes
- Adequate line spacing for comfortable reading

7.3 Responsive Design

The application is fully responsive with:

- Mobile-first design approach
- Breakpoints for tablets and desktops
- Touch-friendly interface elements
- Optimized images and assets

7.4 Accessibility Considerations

Accessibility features include:

- Semantic HTML structure
 - ARIA labels for screen readers
 - Keyboard navigation support
 - Sufficient color contrast ratios
 - Focus indicators for interactive elements
-

8. Setup and Installation

8.1 Prerequisites

Required Software:

- Node.js (v18 or higher)
- npm or yarn package manager
- Git for version control
- Code editor (VS Code recommended)
- Firebase CLI (`npm install -g firebase-tools`)

8.2 Development Environment Setup

1. Clone the Repository:

```
bash
git clone <repository-url>
cd onlyfunds
```

2. Install Dependencies:

```
bash
npm install
```

3. Configure Environment Variables:

Create a `.env.local` file with Firebase configuration:

```
text
NEXT_PUBLIC_FIREBASE_API_KEY=your_api_key
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN=your_auth_domain
NEXT_PUBLIC_FIREBASE_PROJECT_ID=your_project_id
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET=your_storage_bucket
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID=your_sender_id
NEXT_PUBLIC_FIREBASE_APP_ID=your_app_id
```

8.3 Firebase Configuration

1. Create Firebase Project:

- Visit Firebase Console
- Create new project
- Enable required services (Authentication, Firestore, Hosting)

2. Configure Authentication:

- Enable Email/Password provider
- Set up authorized domains

3. Initialize Firestore:

- Create database in production mode
- Deploy security rules

8.4 Running Locally

Start the development server:

```
bash  
npm run dev
```

The application will be available at <http://localhost:3000>.

8.5 Environment Variables

All sensitive configuration should be stored in environment variables and never committed to version control.

9. User Guide

9.1 Getting Started

Visit the OnlyFunds website and click "Sign Up" to create an account.

9.2 Creating an Account

1. Click "Sign Up" button
2. Enter email and password
3. Complete profile information
4. Verify email (if required)
5. Access dashboard

9.3 Setting Up Profile

1. Navigate to profile settings
2. Upload profile picture
3. Add bio and contact information
4. Save changes

9.4 Creating Fund Requests

1. Click "Create Post" button
2. Enter title and description
3. Set funding goal
4. Add images or media
5. Select category
6. Publish post

9.5 Making Donations

1. Browse fund request posts
2. Click "Donate" button
3. Enter donation amount
4. Add optional message
5. Confirm donation

9.6 Interacting with Posts

Users can:

- Like posts to show support
- Comment to ask questions or offer encouragement
- Share posts (future feature)

9.7 Using Chat Feature

1. Search for a user
2. Click on their profile

3. Click "Message" button
 4. Type and send messages
 5. View conversation history
-

10. Developer Guide

10.1 Project Structure

text

```
onlyfunds/
├── public/          # Static assets
├── src/
│   ├── app/        # Next.js pages
│   ├── components/  # React components
│   │   ├── feed/    # Feed-related components
│   │   ├── shared/   # Shared/common components
│   │   └── ui/       # UI primitives
│   ├── lib/         # Utilities and helpers
│   ├── hooks/       # Custom React hooks
│   └── firebase/    # Firebase configuration
├── firestore.rules  # Security rules
├── package.json
└── next.config.js
```

10.2 Code Organization

Components:

- Keep components small and focused
- Use composition over inheritance
- Extract reusable logic into custom hooks

State Management:

- Use React hooks for local state
- Firebase real-time listeners for server state
- Context API for global state (when needed)

10.3 Key Components

Logo Component:

Located at `src/components/shared/Logo.tsx`, displays the application logo.

Create Post Dialog:

Handles fund request creation with form validation and Firebase integration.

Feed Component:

Displays list of fund request posts with infinite scroll.

Chat Interface:

Manages real-time messaging between users.

10.4 Adding New Features

Steps to add a feature:

1. Define requirements and data models
2. Update Firestore security rules if needed
3. Create necessary components
4. Implement Firebase integration
5. Add tests
6. Update documentation

10.5 Testing Strategy

Testing Approach:

- Unit tests for utility functions

- Component tests with React Testing Library
- Integration tests for Firebase operations
- Manual testing for user flows

10.6 Code Style Guidelines

Standards:

- Use TypeScript for type safety
 - Follow ESLint rules
 - Use Prettier for code formatting
 - Write descriptive variable and function names
 - Add comments for complex logic
-

11. Deployment

11.1 Deployment Overview

OnlyFunds is deployed on Firebase Hosting, providing:

- Global CDN distribution
- Automatic SSL certificates
- Custom domain support
- Rollback capabilities

11.2 Firebase Hosting Setup

Initialize Hosting:

```
bash
firebase init hosting
```

Configure **firebase.json**:

```
json
{
  "hosting": {
    "public": "out",
    "ignore": ["firebase.json", "**/.*", "**/node_modules/**"]
  }
}
```

11.3 Billing Configuration

Google Cloud Billing:

Firebase requires a linked billing account for production use. Set up:

1. Google Cloud Billing account
2. Link to Firebase project
3. Configure budget alerts
4. Set up autopay (optional)

Free Tier:

Firebase offers generous free tier limits. Most small to medium applications stay within free tier.

11.4 Publishing the Application

Build and Deploy:

```
bash
npm run build
firebase deploy --only hosting
```

Verify Deployment:

- Check deployment URL
- Test all features
- Monitor for errors

11.5 Custom Domain Setup

Add Custom Domain:

1. Go to Firebase Console > Hosting
 2. Click "Add custom domain"
 3. Enter your domain name
 4. Verify ownership
 5. Update DNS records
 6. Wait for SSL provisioning
-

12. Security and Privacy

12.1 Security Architecture

OnlyFunds employs multiple layers of security:

- Firebase Authentication for user identity
- Firestore Security Rules for data access control
- HTTPS encryption for all communications
- Input validation and sanitization

12.2 Firestore Security Rules

Key Principles:

- Deny by default, allow explicitly
- Authenticate users before granting access
- Validate data on write operations
- Enforce ownership for sensitive data

User Data Protection:

Users can only modify their own profiles. Public read access is limited to non-sensitive fields.

Post Security:

Posts are publicly readable but only the author can modify or delete them.

Chat Privacy:

Chat messages are strictly private between sender and receiver.

12.3 Authentication Security**Password Requirements:**

- Minimum length enforced
- Secure password storage (handled by Firebase)
- Password reset via email verification

Session Management:

- Secure token-based authentication
- Automatic token refresh
- Logout clears local tokens

12.4 Data Privacy**User Privacy:**

- Email addresses are not publicly displayed
- Users control visibility of profile information
- No tracking without consent

Data Collection:

- Minimal data collection
- Clear privacy policy
- User data deletion on request

12.5 Best Practices

For Developers:

- Never commit API keys or secrets
- Use environment variables for configuration
- Regularly update dependencies
- Follow OWASP security guidelines

For Users:

- Use strong, unique passwords
 - Enable two-factor authentication when available
 - Report suspicious activity
 - Keep contact information updated
-

13. Performance and Optimization

13.1 Performance Considerations

Frontend Optimization:

- Code splitting for faster initial load
- Image optimization and lazy loading
- Minification and compression
- Caching strategies

Backend Optimization:

- Efficient Firestore queries
- Proper indexing
- Pagination for large datasets
- Real-time listener optimization

13.2 Database Optimization

Query Optimization:

- Use composite indexes for complex queries
- Limit query results with pagination
- Cache frequently accessed data
- Avoid unnecessary reads

Data Modeling:

- Denormalize data when appropriate
- Use subcollections for scalability
- Keep documents small
- Batch writes when possible

13.3 Frontend Optimization

React Performance:

- Use React.memo for expensive components
- Optimize re-renders with useMemo and useCallback
- Lazy load routes and components
- Virtual scrolling for long lists

Asset Optimization:

- Compress images
- Use next/image for optimized images
- Implement progressive web app features
- Minimize bundle size

13.4 Caching Strategies

Browser Caching:

- Leverage browser cache for static assets
- Use service workers for offline support
- Implement stale-while-revalidate pattern

Firestore Caching:

- Enable Firestore offline persistence
 - Cache authentication state
 - Use Cloud CDN for hosting
-

14. Maintenance and Updates

14.1 Regular Maintenance

Routine Tasks:

- Monitor application health
- Review error logs
- Check performance metrics
- Update dependencies
- Backup database

14.2 Updating Dependencies

Update Process:

1. Check for updates: **npm outdated**
2. Review changelogs for breaking changes
3. Update packages: **npm update**
4. Test thoroughly before deploying
5. Document any breaking changes

14.3 Backup and Recovery

Firestore Backups:

- Automated daily backups via Firebase
- Manual export for critical data
- Test restore procedures regularly

Code Backups:

- Version control with Git
- Regular commits with meaningful messages
- Tag releases for easy rollback

14.4 Monitoring and Analytics

Firebase Analytics:

- User engagement metrics
- Feature usage tracking
- Error monitoring
- Performance monitoring

Custom Monitoring:

- Set up alerts for critical errors
- Monitor database usage
- Track authentication failures
- Review security rule violations

14.5 Error Handling

Client-Side Errors:

- Global error boundary in React

- User-friendly error messages
- Error logging to Firebase

Server-Side Errors:

- Firebase Functions error handling (if used)
 - Proper error responses
 - Retry logic for transient failures
-

15. Troubleshooting

15.1 Common Issues

Build Errors:

- Clear node_modules and reinstall
- Check Node.js version compatibility
- Verify environment variables

Runtime Errors:

- Check browser console for errors
- Verify Firebase configuration
- Check network connectivity

15.2 Firebase Permission Errors

"Missing or insufficient permissions" error:

1. Check Firestore security rules
2. Verify user is authenticated
3. Ensure user has required permissions
4. Check for typos in collection/document paths

Solution:

Update security rules to grant appropriate access while maintaining security.

15.3 Authentication Problems

Login Failures:

- Verify email/password are correct
- Check Firebase Authentication is enabled
- Ensure email provider is enabled
- Review authentication logs in Firebase Console

Session Expiry:

- Implement token refresh logic
- Handle expired session gracefully
- Redirect to login when necessary

15.4 Deployment Issues

Deployment Fails:

- Check Firebase CLI is updated
- Verify Firebase project is selected: `firebase use`
- Ensure build completed successfully
- Check hosting quota limits

Site Not Updating:

- Clear CDN cache
 - Wait for propagation (can take a few minutes)
 - Hard refresh browser (Ctrl+Shift+R)
 - Check deployment logs
-

16. Future Roadmap

16.1 Planned Features

Short-term (3-6 months):

- Payment gateway integration
- Enhanced search filters
- Push notifications
- Email notifications
- User verification system
- Social media sharing

Medium-term (6-12 months):

- Mobile applications (iOS/Android)
- Advanced analytics dashboard
- Automated content moderation
- Multi-language support
- Recurring donations
- Campaign updates feature

Long-term (12+ months):

- AI-powered fund matching
- Blockchain integration for transparency
- Corporate donor portal
- API for third-party integrations
- Advanced reporting tools

16.2 Scalability Improvements

Infrastructure:

- Implement Cloud Functions for backend logic
- Add Algolia for advanced search

- Use Cloud Storage for media files
- Implement Redis for caching

Performance:

- Optimize database queries
- Implement better caching strategies
- Use CDN for static assets
- Optimize image delivery

16.3 Enhanced Analytics

User Analytics:

- Detailed user behavior tracking
- Conversion funnel analysis
- A/B testing framework
- Cohort analysis

Business Metrics:

- Total funds raised
- Success rate of campaigns
- User retention metrics
- Donor lifetime value

16.4 Mobile Application

Native Apps:

- iOS app using Swift
- Android app using Kotlin
- Shared codebase where possible
- Push notification support
- Offline functionality

17. Appendix

17.1 Glossary

Firestore: Google's NoSQL cloud database

Firebase Authentication: User authentication service

Next.js: React framework for web applications

TypeScript: Typed superset of JavaScript

CDN: Content Delivery Network

API: Application Programming Interface

SDK: Software Development Kit

NoSQL: Non-relational database

SSR: Server-Side Rendering

PWA: Progressive Web App

17.2 References

Official Documentation:

- Firebase Documentation: <https://firebase.google.com/docs>
- Next.js Documentation: <https://nextjs.org/docs>
- React Documentation: <https://react.dev>
- TypeScript Documentation: <https://www.typescriptlang.org/docs>

Community Resources:

- Firebase Stack Overflow
- Next.js GitHub Discussions
- React Community Forums

17.3 Change Log

Version 1.0.0 (October 2025):

- Initial release
- Core features implemented
- Firebase integration complete
- Security rules deployed

Future Versions:

- Will be documented here as updates are released

17.4 Contact Information

Development Team:

- Project Lead: G Kushal - gkushal154@gmail.com
- Technical Support: Y Sai Charan – ysaicharan173@gmail.com
- Bug Reports: Y Sreenath – ysreenath188@gmail.com

OnlyFunds

HomeChat

Admin

Filters

Keyword Search

Search by keyword...

Categories

EmergencyCommunityEducationHealthCreativeTravel

Hide Fully Funded

Top Donors

1Prabhas₹4,76,053

2user22₹3,96,280

TFIBanisaRamesh · 1 day ago

Jai Mahishmathi! For the love of cinema.

tfi baagundali

guys nenu bahubali the epic release kosam full waiting naku imax lo chuudalani korika but moeny levu, pls any rebel fans, rajamouli fans help cheyandi, rana fans chesina parledu, jai mahismathi

₹1,500raised of ₹1,500

CreativeCinema

1656

Funded

KiranAbbavaram · 1 day ago

Aspiring filmmaker. Cinema is life.

naku urgent ga oka camera kaavali

hlo guys, naak cinema ante pranam, naku oka short film idea, theesthe utube motham mandhe pls help for camera

₹1,50,000raised of ₹2,00,000

CreativeUrgentCinema

user22

Verified

Cinema is my passion. Here to support inspiring stories.

View Profile

OnlyFunds

HomeChat

Admin

Chats

Search users...

Prabhas

Hey! How's it going?

AlluArjun

Click to start chatting...

NTRJr

Click to start chatting...

RamCharan

Click to start chatting...

kushalcr

Click to start chatting...

KiranAbbavaram

Click to start chatting...

SampurneshBabu

Click to start chatting...

VikramArditva

Click to start chatting...

Prabhas

Online

Hey! How's it going? Thanks so much, I appreciate that!

Hi there! I saw your post, really inspiring.

studio---onlyfunds2-42418579-66f16.us-central1.hosted.app/chat

OnlyFunds

HomeChat

Admin

user22

Verified

Cinema is my passion. Here to support inspiring stories.

View Profile

Posts by user22

No posts found

It looks like there are no posts yet. Why not create one?

OnlyFunds

Home

Chat

Admin

Admin Panel

Content Moderation

Reported Content

User Management

Audit Log

AI Content Moderation

Enter content to check if it meets platform guidelines. The AI will flag inappropriate content based on its category and tags.

Content

Enter post or comment text here...

Category

e.g., Emergency

Tags (comma-separated)

e.g., Urgent, Accident

Moderate Content

OnlyFunds

Home

Chat

Admin

Filters

Keyword Search

Search by keyword...

Categories

EmergencyCommunity

EducationHealthCreative

Travel

Hide Fully Funded

Top Donors

1

NTRJr

₹3,24,956

2

user22

₹2,53,877

3

DevChavan

₹1,23,456

Create a New Fund Request

Share your story and what you need funding for. Be detailed and clear.

Request Title

e.g., Emergency Vet Bill for Sparky

Description

Tell your story in detail...

Amount Requested (₹)

400000

Category

Select a category

Tags

Urgent, Medical, Family (comma-separated)

Separate tags with commas.

Media (Optional)

user22

Verified

Cinema is my passion. Here to support inspiring stories.

View Profile

OnlyFunds

Home

Chat

Admin

AlluArjun · 3 days ago

Pushpa means flower? No, Pushpa means fire!

Fund My First Semester of College!

I've been accepted into my dream university but I'm short on funds for the first semester's tuition and books. I'm the first in my family to go to college. The total needed is ₹6,50,000.

₹6,00,000

raised of ₹6,50,000

EducationFamilyCollege

724

₹ Donate

NTRJr · 4 days ago

Man of Masses. Making a difference is the real heroism.

Help Rebuild Our Community Garden

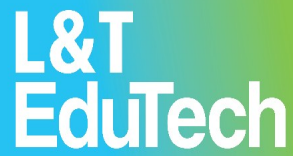
Our local community garden was destroyed in a recent storm. We need funds to buy new soil, seeds, and tools to get it back up and running. It's a vital source of fresh produce for many families.

user22

Verified

Cinema is my passion. Here to support inspiring stories.

View Profile



CERTIFICATE OF COURSE PATHWAY COMPLETION

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has completed the Course pathway titled
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JavaScript**
which covers
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CID: LTE/EI/1000

M.F.Febin

Head
L&T EduTech



**L&T
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